

Restaurant Spending Analysis

Plotly Express tips dataset | Python | Quantitative Analytics

1. Objective

Examine how restaurant bill size varies by day, smoking status and meal period, and test whether party size is associated with total bill value.

2. Dataset and methods

The project uses the Plotly Express tips dataset containing 244 restaurant bills. Variables include total bill, tip, sex, smoker, day, time and party size. Analysis is conducted in Python using pandas, SciPy and statsmodels.

3. Statistical workflow

The analysis calculates descriptive statistics by day, smoking status and meal period; applies a Welch independent-samples t-test comparing smokers and non-smokers; constructs low/medium/high bill categories for a Chi-square analysis by day; and estimates a simple OLS regression of total bill on party size.

4. Assumption and sensitivity checks

Normality is assessed using Shapiro-Wilk tests, variance equality using Levene's test, and regression residuals using Shapiro-Wilk and Breusch-Pagan diagnostics. HC3 robust standard errors are also calculated as a sensitivity check.

5. Business interpretation

Party size is a natural operational driver of bill value and can inform table allocation, staffing and capacity planning. Differences by day, smoking status or meal period can help managers understand demand patterns. These relationships are observational and should not be interpreted as causal without a stronger design.

6. Reproducibility

The repository contains analysis.py, the tips.csv dataset, supporting visualisations and this report. The script can be run from the project directory after installing pandas, numpy, scipy and statsmodels.